

Paper 25.75 - Energetic Traversal Next-State Precondition

CQE-paper-25.75

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-13

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-25.75.md

Paper 25.75 - Energetic Traversal Next-State Precondition

Purpose

This supplement defines what Paper 25 exports to the next state of the suite. It is a precondition layer, not a new proof claim.

Exported Preconditions

Any downstream paper that uses traversal cost must inherit:

- one NSL row per step,
- additive ``theta_path``,
- explicit unit policy,
- obligation carry for positive or uncalibrated rows,
- default VOA analog cost when the row is still at chart-state level,
- refusal to promote analog accounting to physical energy without calibration.

Downstream Use

Paper 26 and later horizon papers may use the energetic traversal ledger to compare routes, detect boundary collisions, and decide where a new verifier must attach. They must not treat the ledger as a replacement for the domain's own metric or measurement.

Kernel Handshake

The kernel sidecar should expose Paper 25 as a boundary adapter: receive a transport row, compute ``theta``, return closure or obligation, and preserve the unit policy. In training mode, the guess label stays hidden until after the agent or reviewer predicts whether the row closes.